



Waymo Technical Screen: ML Coding & Problem Solving

Our team is looking forward to connecting with you soon! Below you'll find a breakdown of what to expect during your interview, preparation resources, and tips for success. This is not intended to be an exhaustive list, but it should give you a general idea of the focus areas we cover.

Interview Overview

Introduction

You'll spend the first 5 minutes or so getting to know each other. You can expect us to ask about:

- Your experience: a project you've worked on and your contributions, or something interesting on your resume
- Your motivation: why Waymo/this role, which technologies interest you

Coding

The next 30 minutes will involve a coding exercise via [CoderPad](#) in the language discussed with your Sourcer or Recruiter (typically Python using NumPy or PyTorch). Overall, we're looking for candidates who are great problem solvers, quick learners, and clear communicators. Ideal solutions will make significant progress through the technical problem with clean, efficient, and "tensorized" code. We look for the ability to use broadcasting and vectorized operations properly rather than writing slow loops. Be sure to test your code, recognize failure modes (e.g., numerical instability, overfitting), and strive for optimized solutions in terms of computational efficiency.

With these general guidelines in mind, please note that different coding questions test for different things. Some may look for a complete implementation of a specific algorithm (e.g., K-Means), while others evaluate your ability to formulate an ML problem from scratch. Discuss expectations with your interviewer before diving in to make sure you focus on the right objective.

Practice & Preparation

We recommend practicing implementing basic ML algorithms to efficiently solve problems, know their mathematical foundations, common operations, and when to use them.

- **Supervised Learning:** Implementations of Linear/Logistic Regression (gradient descent, cost functions), Decision Trees (entropy, pruning), and KNN (distance metrics).
- **Unsupervised Learning:** K-Means Clustering (centroids) and PCA (dimensionality reduction, eigenvectors).
- **Neural Networks:** Feedforward architectures, backpropagation, activation functions, and Softmax.
- **Deep Learning Operations:** Convolutions, pooling layers, and sequence processing (RNNs/LSTMs).
- **Sampling & Math:** Stratified vs. Uniform sampling, reservoir sampling, & generating data from distributions.
- **Optimization:** Understanding SGD, Momentum, and Adam optimizers.

Suggested Questions to Practice:

- [Implement Linear Regression from scratch](#) (using NumPy)
- [Implement Logistic Regression and a custom cost function](#)
- [K-Nearest Neighbors](#) (focus on vectorization/broadcasting) ([Example 2](#))
- [Implement a Decision Tree Classifier](#) (calculating Gini impurity)
- [Implement Softmax and Cross-Entropy Loss](#)
- [Convolution operation from scratch](#) (handling padding/strides)
- [Tensor Puzzles](#) (broadcasting exercises)

Additional Resources:

[Google Machine Learning Crash Course](#)

[ML System Design Questions](#)

[Tensor Gym](#) - Interactive exercises specifically for practicing tensor operations and broadcasting.

How to succeed

At Waymo, we believe in collaboration and sharing ideas. Most importantly, you'll need more information from the interviewer to analyze and answer the question to its full extent.

- **Ask Clarifying Questions:** We ask open-ended questions to learn how you approach and solve problems. You might be asked to "build a classifier for X." Your interviewers are looking for you to narrow this down: Is it multi-class? What are the input features? How do we evaluate success?
- **Define and Frame the Problem:** Start by defining the ML task concretely. If the problem is broad, break it down: discuss data sources, feature engineering, model selection, and evaluation metrics.
- **Show Your Thinking:** As you solve the problem, communicate your decision-making process. Discuss potential trade-offs (e.g., complex model vs. inference speed), pitfalls (e.g., class imbalance, missing data), and edge cases. We want to understand how you use data to inform decisions.
- **Refine Your Solution:** The first answer that comes to mind may need refining. You might start with a brute-force loop, but a strong candidate will refine this into a vectorized solution using tensors. Think about how to improve the efficiency and how you would verify the model is learning (e.g., checking loss convergence).
- **Connecting the Dots:** When considering which examples to highlight from your past work, find examples where you dealt with ambiguity in data, modeling complexity, or deploying models in production. These experiences help showcase your ability to thrive in our working environment.

What is Waymo looking for?

Our interviewers will be looking at the approach to questions as much as the answer.

- Does the candidate listen carefully and comprehend the ML constraints?
- Does the candidate ask appropriate clarifying questions (e.g., regarding data shape or distribution) before proceeding?
- Does the candidate state their assumptions and incorporate feedback?
- Does the candidate demonstrate comfort with vectorization and broadcasting over brute force loops?
- Are hints heard and heeded?
- Are new ideas and methods of tackling a problem sought?
- Is the candidate inventive and flexible in their solutions (e.g., suggesting alternative architectures)?
- Can questioning move up to more complex problem solving (e.g., handling scale or noise)?

Questions & Wrap Up

At the end of the interview, we recommend having a few questions ready so you can learn more about your potential new team/colleague, work environment, company culture, etc.

Thank you for taking the time to interview with Waymo. Please let your recruiter or coordinator know if there's anything else we can do to help you feel comfortable and successful!